C++14 Fundamentals - Student Edition

Week 4 - Class 7: Operators and Conversions

Why This Matters

Operators allow arithmetic, comparison, logic, and assignment. Conversions control what happens with mixed types. Understanding precedence, division rules, and promotions is vital to avoid bugs.

Arithmetic Operators

Supported: + - * / %

In C++ int division truncates toward 0, not like Python's floor division.

```
#include <iostream>
using namespace std;
int main(){
    cout << 7/2 << "\n";   // 3
    cout << -7/2 << "\n";  // -3, truncates toward 0
}
```

Floor Division vs Truncation

In Python, // is floor division (rounds down): -7//2 = -4.

In C++, integer division truncates toward zero: -7/2 = -3.

Important difference.

Comparison and Logical Operators

== != < <= > >=

&& || !

Assignment and Compound

=, +=, -=, *=, /=, %= etc.

Operator Precedence

Simplified:

()

* / %

+ -

<< >>

< <= > >=

== !=

&&

||

= += -= ...

Implicit Conversions (Hierarchy)

Hierarchy (highest to lowest):

1. long double
2. double
3. float
4. unsigned long long
5. long long
6. unsigned long
7. long
8. unsigned int
9. int
10. char, bool, short

Examples: int+double -> double.

short in expressions -> int.

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Explicit Conversions

Use static_cast<T> for clarity.
Example: double pi=3.14; int n=static_cast<int>(pi);

Common Mistakes

1. Integer division surprises (7/2=3).
2. Signed/unsigned mix-ups (-1<1u).
3. Forgetting precedence.
4. Using C-style casts.

Exercises

1. Compute 7/2 with int and double.
2. Test -7/2 in C++ vs Python -7//2.
3. Predict 3+4*5 vs (3+4)*5.
4. Mix signed and unsigned.
5. Use static_cast to convert double to int.

Project Tie-In

Add to Type Explorer: show int vs double division, compare -7/2 vs Python -7//2, print mixed-type results showing promotions.